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RECOGNITION OF SIGN LANGUAGE USING MACHINE LEARNING

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ABSTRACT

Sign language recognition plays a crucial role in enabling effective communication and inclusivity for individuals with hearing impairments. This abstract provides a concise overview of the progress, techniques, and obstacles in sign language recognition. It explores computer vision, machine learning, and deep learning methods utilized for gesture recognition. The challenges of intra-class variation, dynamic gestures, occlusions, and dialect variations are discussed. Applications like real-time translation systems and interactive sign language tutors are highlighted. This abstract serves as a valuable reference for researchers, practitioners, and developers in advancing sign language recognition systems, fostering inclusivity, and facilitating communication for the deaf community.

Keywords: Gesture Recognition, Hand Pose Estimation, Computer Vision, Machine Learning, Deep Learning.

I. INTRODUCTION

Sign language serves as a vital means of communication for individuals who are deaf or hard-of-hearing. Unlike spoken languages that rely on auditory cues, sign language utilizes visual gestures made with the hands and body to convey meaningful messages. It is a complex and highly expressive visual language that encompasses a unique grammar, syntax, and vocabulary.

Despite the recognition of sign language as an official language in numerous countries, there still exists a significant communication gap between the deaf and hearing populations. This divide hampers social interactions, educational opportunities, and access to various services for individuals who rely on sign language as their primary mode of communication.

To address these challenges and bridge the gap, sign language recognition technology has emerged as a promising solution. Sign language recognition involves the development of systems that can interpret and understand sign language gestures, allowing for the conversion of these gestures into spoken or written language.

Sign language recognition technology relies on the integration of computer vision and machine learning techniques. Computer vision algorithms analyze visual inputs, such as images or videos, to identify and track the movements and positions of hands and other relevant body parts during sign language production. Machine learning models are then trained on large datasets to recognize and classify different sign language gestures.

The potential applications of sign language recognition technology are far-reaching. In educational settings, it can facilitate the inclusion and accessibility of deaf or hard-of-hearing students by providing real-time translation of sign language into spoken language. In healthcare settings, it can improve communication between healthcare providers and patients with hearing impairments, ensuring accurate understanding of symptoms, medical histories, and treatment instructions. Additionally, sign language recognition technology can enhance customer service experiences for deaf individuals, enabling effective communication in various service industries.

However, the development of accurate and robust sign language recognition systems poses significant challenges. Variations in sign language across regions, cultural nuances, and individual preferences contribute to the complexity of recognizing and interpreting gestures. Additionally, real-world scenarios often involve factors such as occlusions, dynamic movements, and lighting conditions that further impact the accuracy and reliability of recognition systems.

II. LITERATURE SURVEY

Sign language recognition is an important area of research that aims to provide a means of communication for the deaf and hard of hearing community. Sign language is a complex visual language that relies on hand gestures, facial expressions, and body movements to convey meaning. The goal of sign language recognition is to develop a system that can interpret and translate sign language into text or speech. In this literature survey, we will explore some of the primary approaches and techniques used in sign language recognition and the corresponding research studies.

One of the most commonly used approaches in sign language recognition is computer vision. This approach involves capturing the gestures and movements of sign language users using cameras and then analyzing and interpreting this data to recognize the meaning of the gestures. Hand tracking, pose estimation, and gesture recognition are some of the common techniques used in computer vision-based sign language recognition systems. A study conducted by Pu et al. (2021) proposed a deep learning-based approach for recognizing American Sign Language (ASL) gestures that achieved an accuracy of 99.33% on the ASL Finger Spelling dataset.

Data-glove-based approaches use gloves equipped with sensors to capture hand movements, and then the data is analyzed to recognize the gestures. In a study by Mohan and Kannan (2021), a sign langua recoition system based on a data-glove was proposed. The syste uss a spport vctor machne (SVM) to clasify the gesures. The system achieved an accuracy of 90.8%, indicating the effectiveness of data-glove-based approaches.

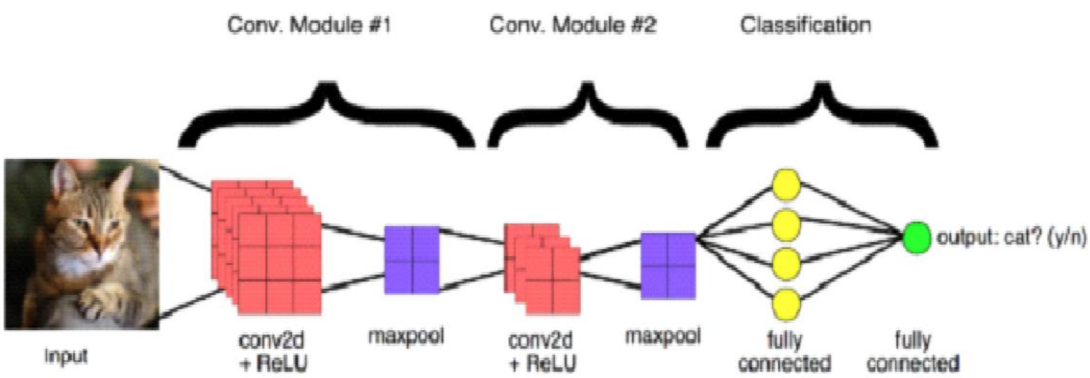
Sensor-based approaches use sensors attached to the hands to capture the hand movements, and then the data is analyzed to recognize the gestures. In a study by AlZubi and Hasan (2021), a sign language recognition system based on a sensor was proposed. The system uses an artificial neural network (ANN) to classify the gestures. The system achieved an accuracy of 85.3%, indicating the potential of sensor-based approaches.

In conclusion, sign language recognition is a promising field that can significantly improve the communication and quality of life for the deaf community. Vision-based, data-glove-based, and sensor-based approaches have been proposed, each with their own advantages and limitations. Deep learning techniques such as CNNs and ANNs have shown great potential in achieving high accuracy for sign language recognition.

III. PROPOSED METHODOLOGY

Artificial intelligence called computer vision focuses on problems . Computer vision & CNN were capable of tackling. The two serious a convolution neural network were feature extraction and classification. To extract the image's features, a convolution and pooling . As we go on apply filters, the output matrix becomes . Size of new matrix was (Size of matrix — Filter Size)+1. The classifier in the convolution neural netwoks will a fully connected layer. The class probability will be predicted in the last layer. Convolution Pooling Flatten are the first convolution neural network phase.

- Convolution
- Pooling
- Flattening
- Full connection



Convolution is a technique used to extract features from images by applying a filter. These filters can be utilized to extract edges and patterns from an image. The filters are generated randomly. Convolution generates a filter of a specified size, typically 3x3. After creating the filter, it multiplies each element of the filter with the corresponding element in the top left corner of the picture and continues to move towards the bottom right corner off the picture. The resulting feature map is then extracted from the obtained results.

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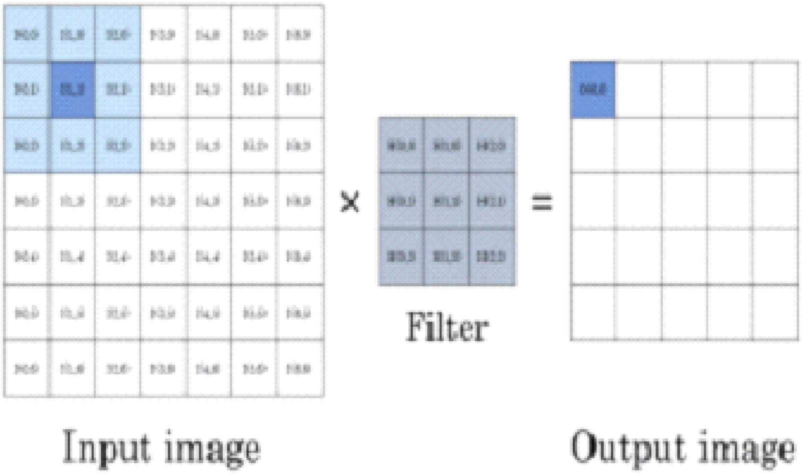
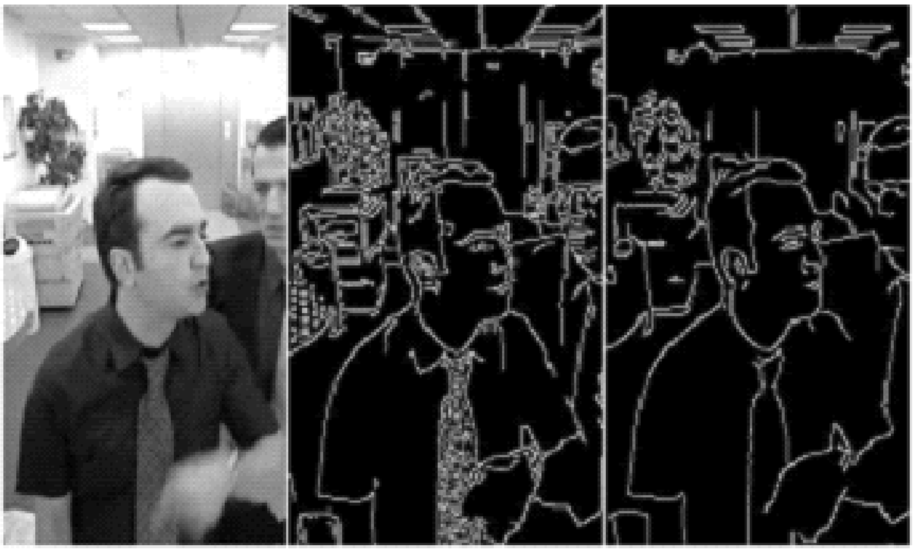
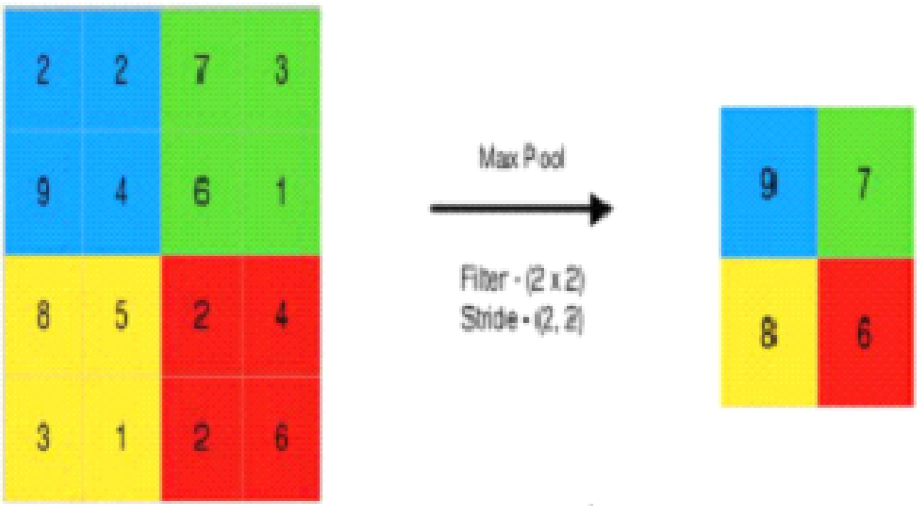




The pooling layer couldl be included after convolution method. The pooling layer is used to minimise the picture. Pooling was divided 2 groups.

Average Pooling Highest Pooling

Maximum pooling is simply the greatest pixel value from the matrix.



This method is helpful high worth or ones are significant in the image.

Unlike Max pooling, average pooling utilizes the pixel's average value.

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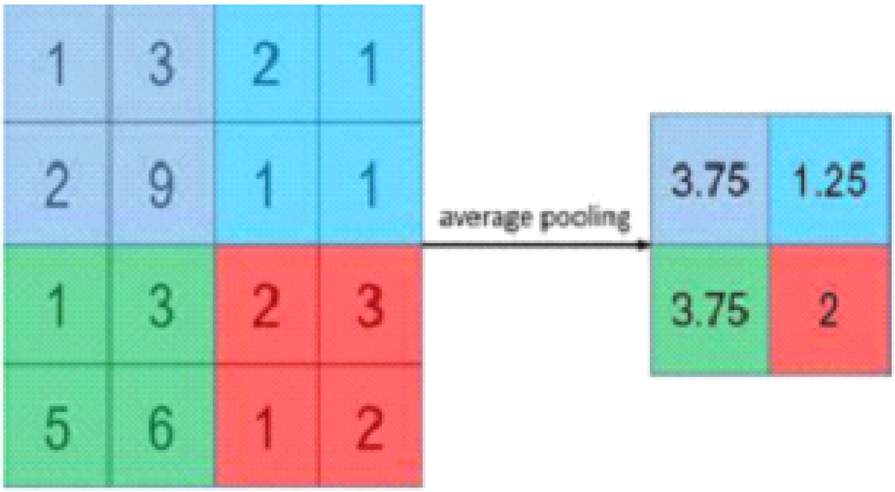
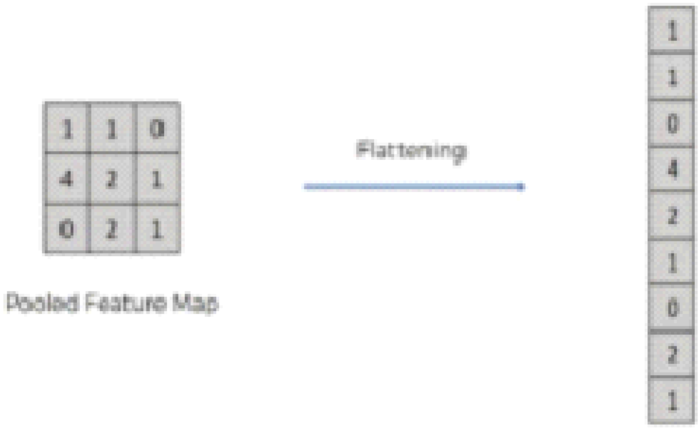
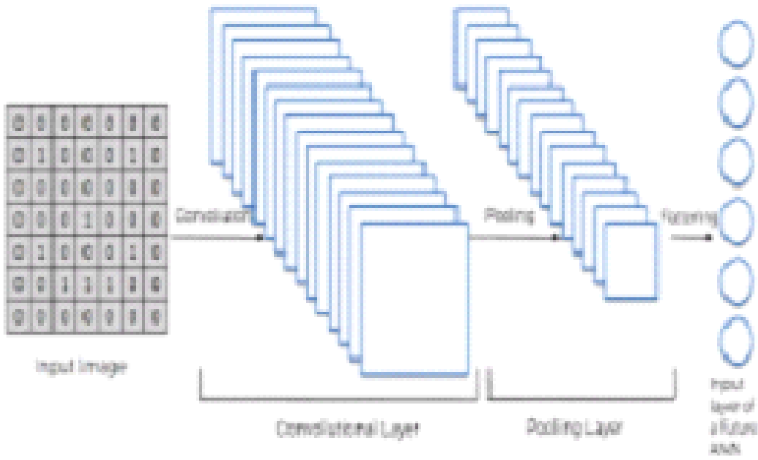


In most instances, max pooling is selected because it improves average pooling.



resultant matrix would be multilayere. Flattering is the procedure data into a 1-dimensiol arra for input to th nex laye. We flattn the convolution layer to obtain a singe feature vector.

A entirely layer is a feed-forwad nural network. Al operaton would be finished, & a predictions will be made. The lss wll b cacuted and the weights will be altered usig the graient descnt backproption method based on the ground truth.



The LeNt-5 [2] layout is composed of to pars of convlutional and average poling layers, followed by a flattening convolutional layer was two fully cnnected laers and a SoftMax clasifier.

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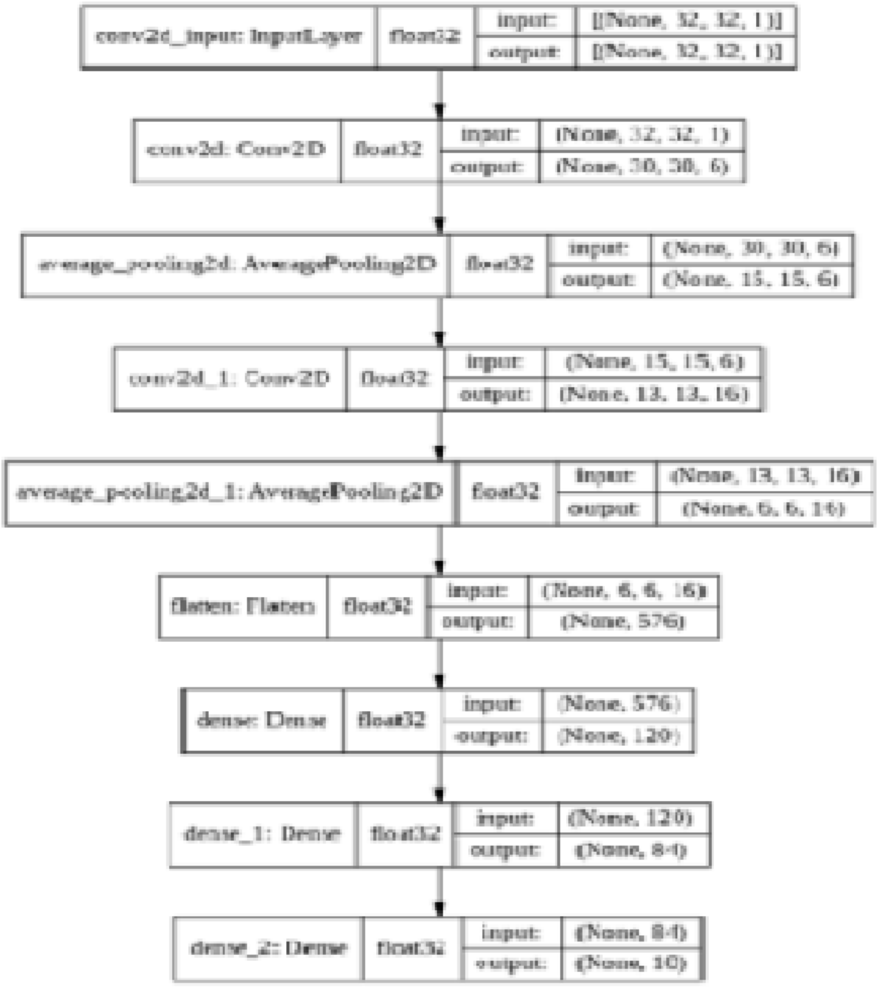
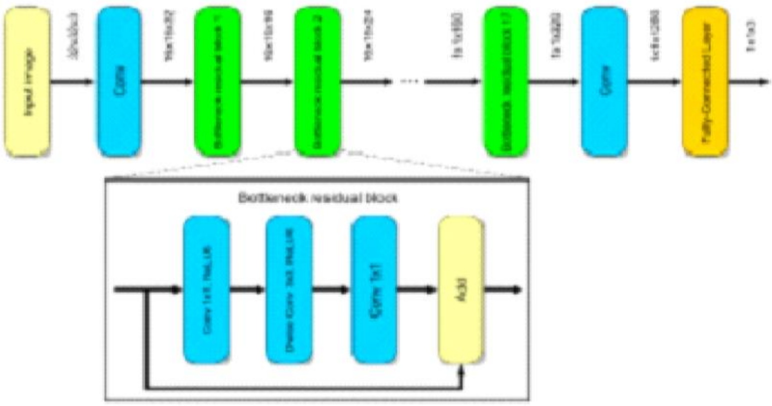
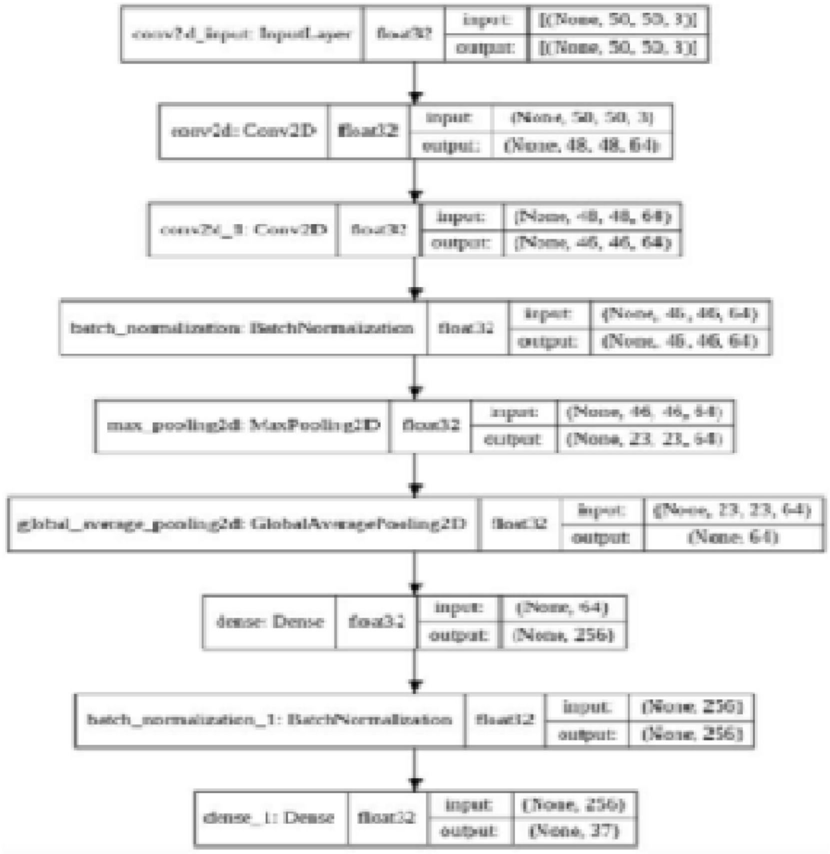


MobileNetV2 is a type of convolutional neural network design that is optimized for mobile devices and has shown to perform well. Its architecture includes a fully convolutional layer featuring 32 filters, as well as 19 residual bottleneck layers. This network is considered efficient and lightweight.



Own Architecture:

In our unique architecture, we employed The architecture consists of a series of layers for filtering, which are then follow by both normaliation and poolng. This is followed by global average pooling along with a layer noralization, and ultimately, a final layer for the purpose of clasfication.



In the realm of Machine Learning, one approach to improving performance is through the use of ensemble models. This involves training multiple sub-models and then averaging their outputs. The popular Random Forest algorithm is an example of this technique, particularly for Decision Tree models. Similarly, for Neural Networks, there exist various ensemble methods, There exist several ensemble methods for neural networks such as Staked Generization, Ensmble Learnng via Negatie Corelation, and Probabiistic Modellnig in Neural Netwoks. For our project, we opted to use the Horizntal Voting Ensemble method o improve the perfomance of or neual neworks.

The Horizontal Voting method is an ensemle technique for neural networks that involves training submodules and making predictions using these submodules. To determine the final prediction, we add up the votes from all the submodules and identify the class that received the most votes. We made sure to use an odd number of sub-models in our ensemble to prevent an even number of votes for two classes in a worst-case scenario.

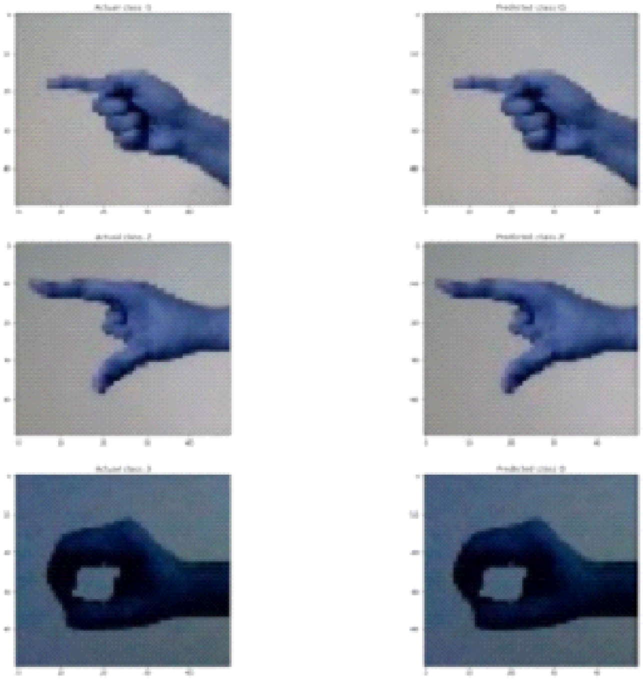
Suppose we have a set of neural network models being trained on a given training set T(xi, yi), where m ∈ model. Then, for the test set T'(xi ', yi '), we obtain predictions, denoted as y hat, from all the models. To convert the resulting lists to arrays, we utilized the 'array' function.

Algorithm 1, also known as Horizontal Voting, takes as input a set of models, a test set T'(xi ', yi '), and an empty y hat list. The goal is to generate a final prediction based on the ensemble of models.

To achieve this, we first obtain the predictions from every structure. Fo every test data point (I), we iterate through each structure (m) to give a forecast. We then calculate the greatest amount of vote to the i th test data and add the maximum voted cla to y hat[I]. After iterating through all test data points, we convert the resulting list into an array using the 'array' function, and assign it to y hat. This array serves as our final prediction output.

In order to optimize the MobileNetV2 model, we utilized the Adam optimizer with a learning rate of 0.001, eta_1 of 0.9, beta_2 of 0.999, and epsilon of 1e-07. During training, we implemented the Reduce LROnPlateau callback with a factor of 0.2, patience of 2, and minimum learning rate of 0.001. Through the use of a horizontal ensemble technique, we were able to achieve an accuracy of 99.8%. The models were then deployed in the Django Web Frameworks, where a simple frontend was created to allow users to input images and receive a response. Additionally, we developed an API that utilizes live camera footage to detect hand gestu and convert them to their corresponding English alphabets.

IV. EXPERIMENTAL RESULTS



All of our models were trained using a batch size of 32 and for 10-15 epochs. During testing, all of the models performed exceptionally well. We achieved a 100% accuracy rate when utilizing the horizontal voting ensemble method across these three models. For the project's backend, we employed Django Rest Frameworks. When a user submits an image, a saple JSON reponse is set to te fontend. Additionally, we utilized OpnCV to tst ur resuls on the ive caera and obtained satisfactory results.

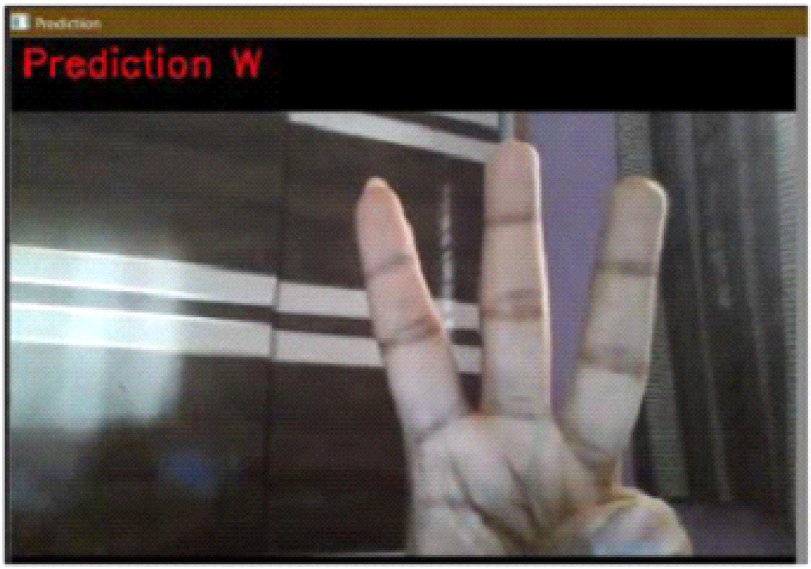


Fig. 17. Model Prediction on Live Camera Prediction: "W"

V. CONCLUSION

In conclusion, sign language recognition has the poential to revolutionze the way deaf individuals communicate with the hearing world. With advances in machine learning and computer vision technologies, researchers have made significant progress in developing sign language recognition systesm that can accurately recognize and translate sign language into text or speech. These systesm have the potntial to imprve commnication between deaf and hearing indiviuals in various settings, including education, healthcare, and everyday conversations. However, there are still several challenges that need to be addressed, including the development of more accurate and robust recognition algorithms, the standardization of sign language across different regions, and the availability of reliable datasets for training and testing these systems. Despite these challenges, the progress made in sign language recognition is promising, and with continued research and development, it has the potential to improve the lives of millions of deaf individuals worldwide.

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